



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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Executive Summary

This project was developed for Space Y in their aim to enter the commercial travel market. A key factor for a successful business model is the landing of the first stage of the rockets in order to use it in other flights. To understand the factors that influence the successful landing of the first stage, a Machine Learning model was built using public data from Space X. The data was collected through APIs and Web Scraping. Data Wrangling was performed to prepare the data for machine learning. Exploratory Data Analysis was performed with SQL and with Data Visualization to identify relationships between the features and identify which ones could be used in the models.

Logistic Regression, Support Vector Machine (SVM), Decision Tree and K Nearest Neighbor (KNN) Machine Learning models were built, optimized and evaluated to select the best model for the purpose.

The SVM model resulted the most appropriate for this task with a precision of 0.845 the success of future launches.

Introduction

Space Y aims to enter the commercial space travel market. To be able to have a successful business, they need to compete with the launch costs promoted by Space X. Since space X low prices lay on the capacity to reuse the first stage, this job consists on understanding what factors determine that the first stage of space X launches lands successfully. For this a ML model will be trained to determine the first stage recovery for each launch using public data from Space X this information will help determine the price of each launch.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Gathered data from SpaceX public API and by scrapping SpaceX Wikipedia page
- Perform data wrangling
 - Filtering the data, dealing with missing values and preparing data for binary classification.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Building, tuning and evaluating several classification models

Data Collection

The data collection process involved a combination of API requests from Space X public API and web scrapping from a table in Space X's entry in Wikipedia.

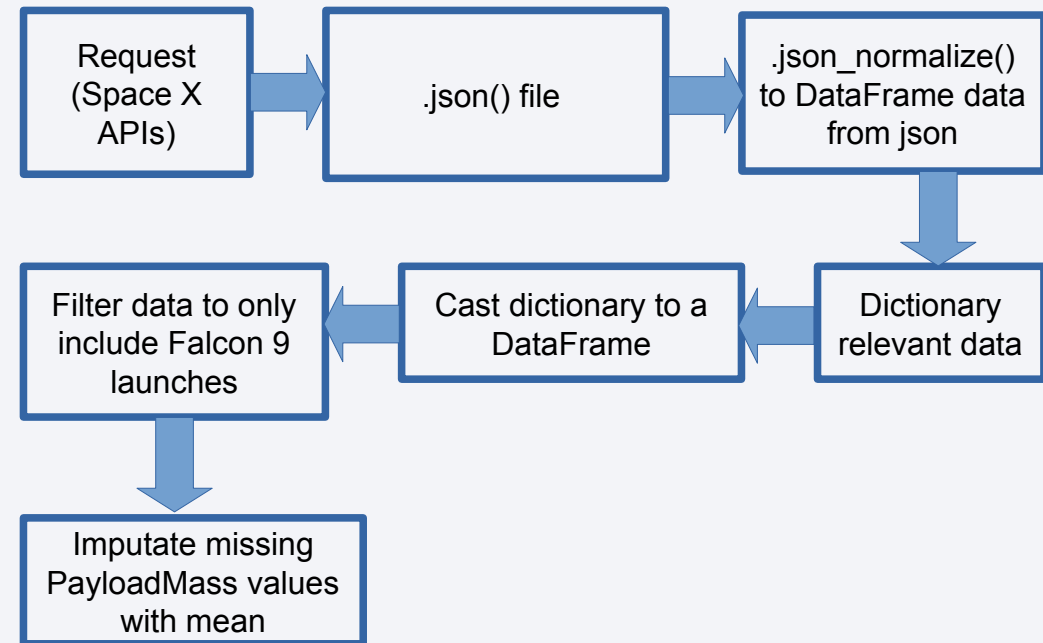
This allowed to obtain a complete information about the launches (payload delivered, launch specifications, landing specifications and landing outcome) for a detailed analysis.

Data Collection – SpaceX API

- Used get request to collect data, cleaned the requested data and did basic data wrangling and formatting

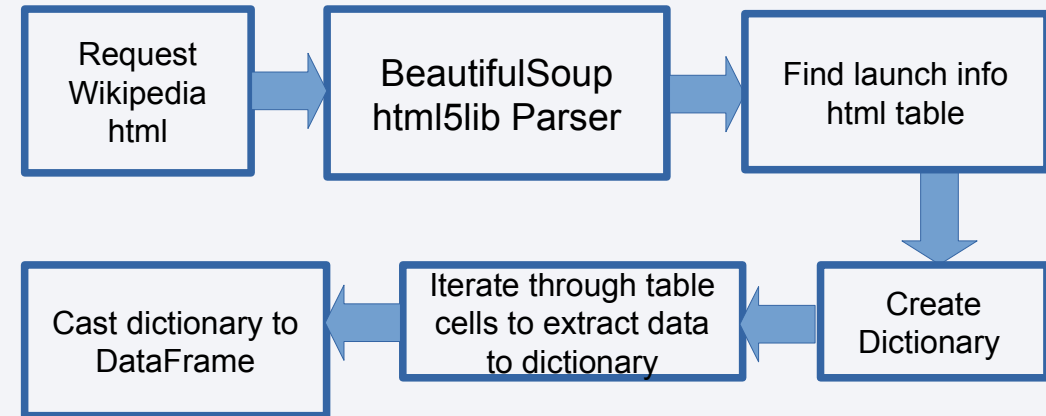
GitHub URL:

https://github.com/morca76/DS_CapstoneProject/blob/main/jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

- Web scrapped Falcon 9 launch records with BeautifulSoup
- Parsed the table and converted it into a pandas dataframe



GitHub URL:

https://github.com/morca76/D_S_CapstoneProject/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling

Created a landing outcome label for each launch:

Successful = 1

Failure = 0.

The mission outcomes were classified as:

True ASDS, True RTLS, & True Ocean – set to -> 1

None None, False ASDS, None ASDS, False Ocean, False RTLS – set to -> 0

GitHub URL:

https://github.com/morca76/DS_CapstoneProject/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

Exploratory Data Analysis was performed on the variables Flight Number, Payload Mass, Launch Site, Orbit, Class and Year to try to find relationships between them, and decide whether to use them to train the machine learning model.

Some data plotted were: Flight Number vs. Payload Mass, Flight Number vs. Launch Site, Payload Mass vs. Launch Site, Success Rate vs. Orbit Type, Orbit vs. Flight Number and Success Yearly Trend.

GitHub URL:

https://github.com/morca76/DS_CapstoneProject/blob/main/edadataviz.ipynb

EDA with SQL

The dataset was loaded into a SQL database and EDA with SQL was performed to get insight from the data. Information obtained:

- Names of the unique launch sites in the space mission.
- Total payload mass carried by boosters launched by NASA (CRS).
- Average payload mass carried by booster version F9 v1.1.
- Total number of successful and failure mission outcomes.
- Ranked counting of landing outcomes.

GitHub URL:

[https://github.com/morca76/DS_CapstoneProject/blob/main/jupyter-labs-eda-sql-coursera sqlite.ipynb](https://github.com/morca76/DS_CapstoneProject/blob/main/jupyter-labs-eda-sql-coursera%20sqlite.ipynb)

Build an Interactive Map with Folium

Maps were created with Folium to visualize the location of the Launch sites, the number of successful and unsuccessful landings per location; and the proximity of the launch sites to locations like railways, highways, coast lines and cities.

This helps to understand why the locations of the launch sites were selected.

GitHub URL:

https://github.com/morca76/DS_CapstoneProject/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

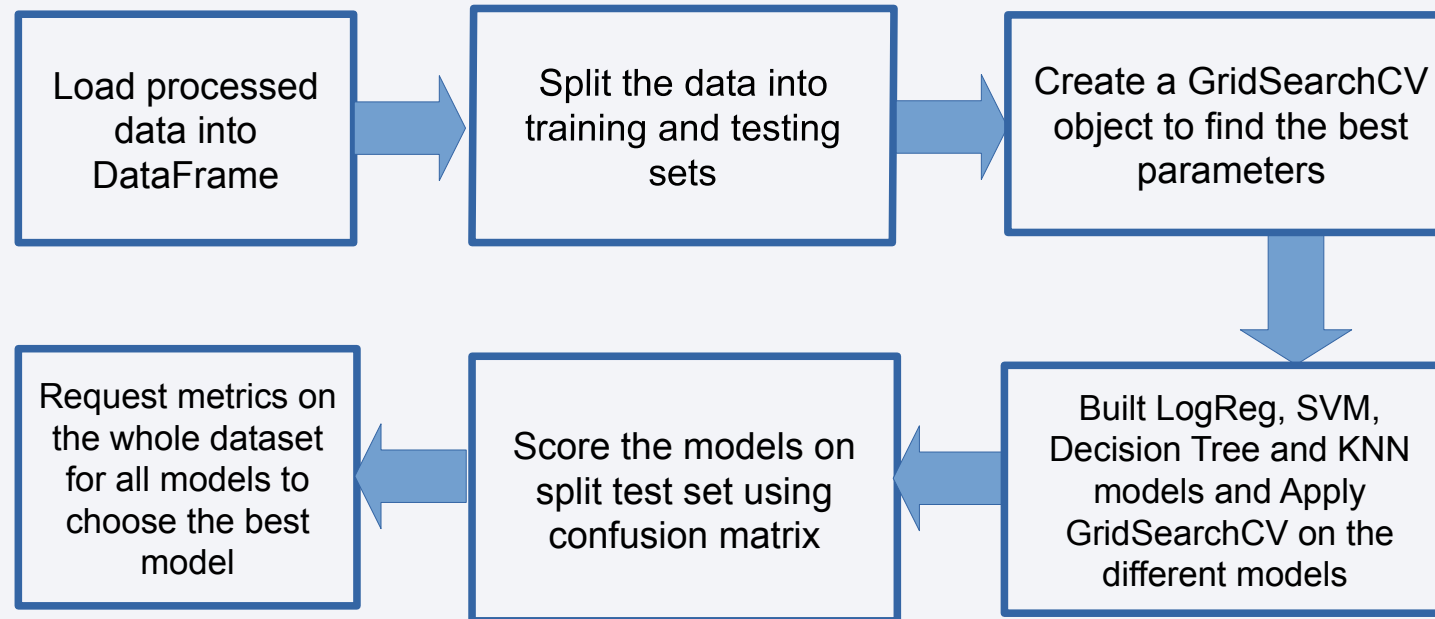
A dashboard with interactive plots was constructed.

It allowed to visualize the contribution to successful launches per launch site. For each site it was visualized the percentage of successful and unsuccessful launches. And both for all launch sites and individual launch sites it was observed the classification of each launch (successful or unsuccessful) per payload mass.

GitHub URL:

https://github.com/morca76/DS_CapstoneProject/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)

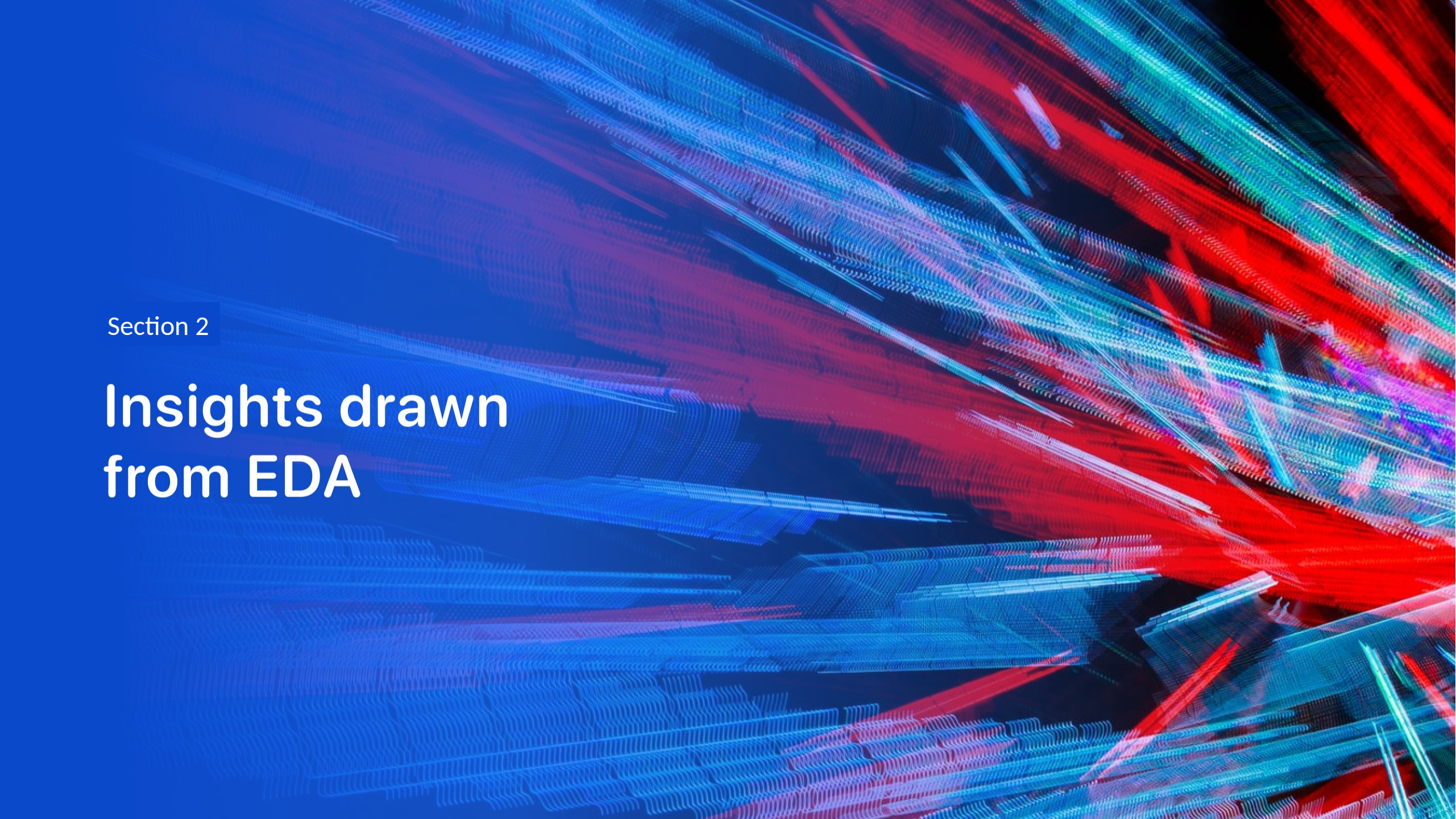


GitHub URL:

https://github.com/morca76/DS_CapstoneProject/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

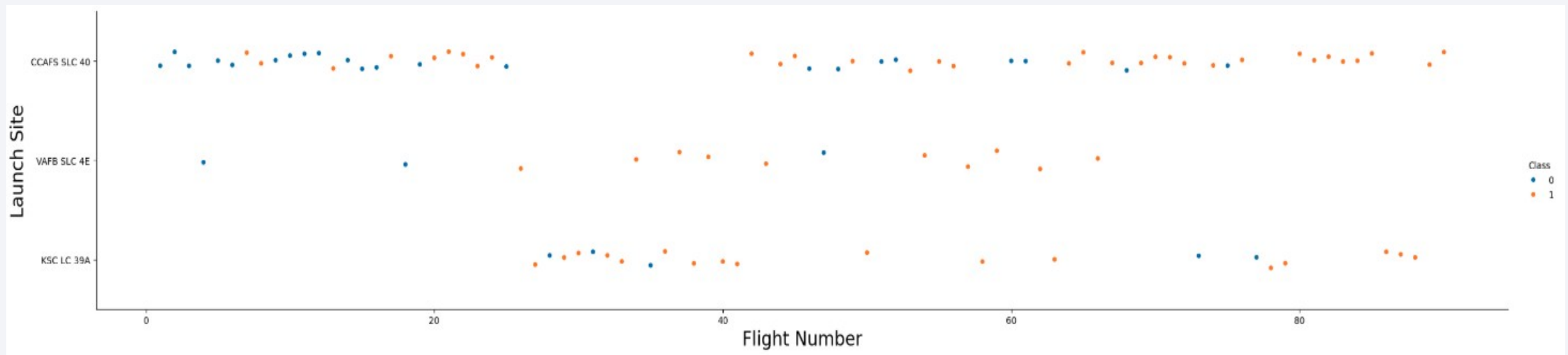
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

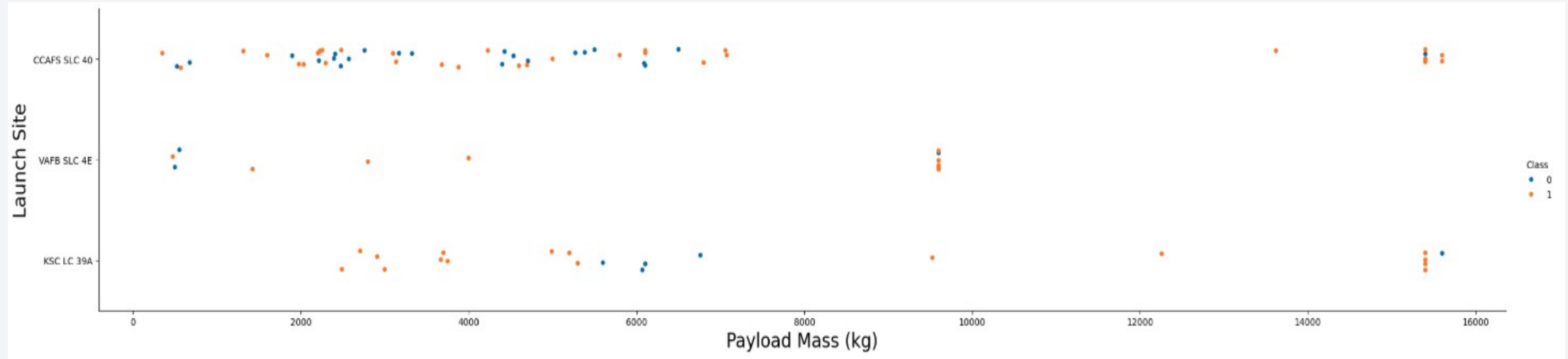
Insights drawn from EDA

Flight Number vs. Launch Site



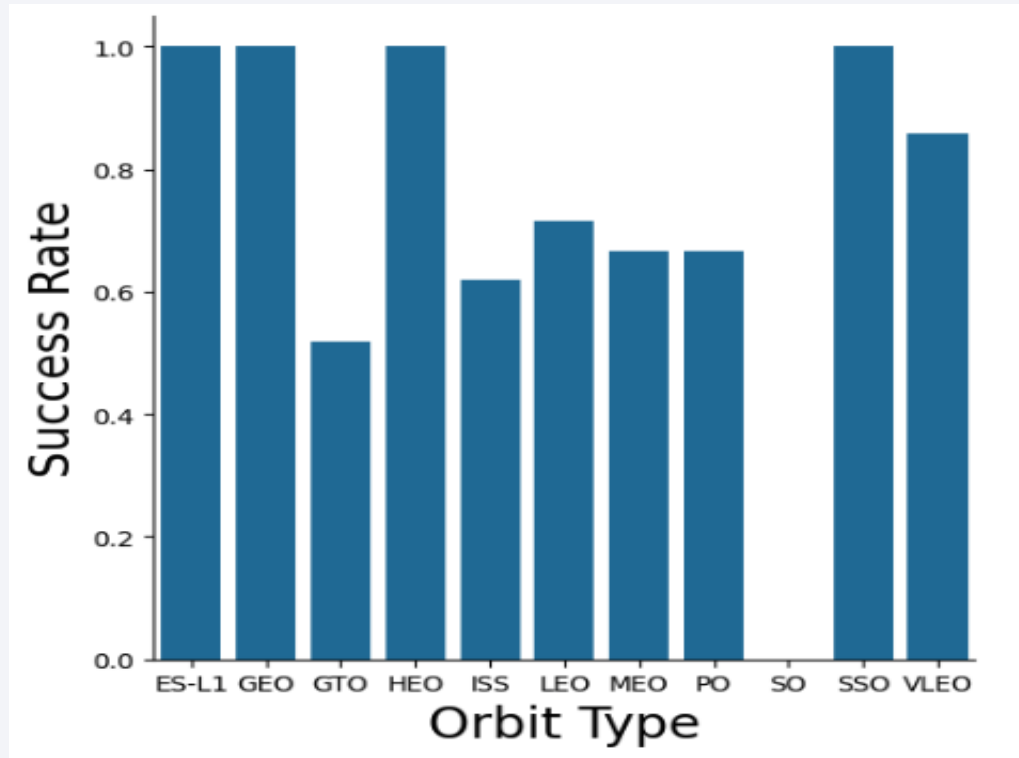
- Most of the earliest flights failed, while most of the latest flights succeeded.
- The CCAFS SLC 40 launch site has the largest number of launches.
- VAFB SLC 4E and KSC LC 39A have higher success rate.

Payload vs. Launch Site



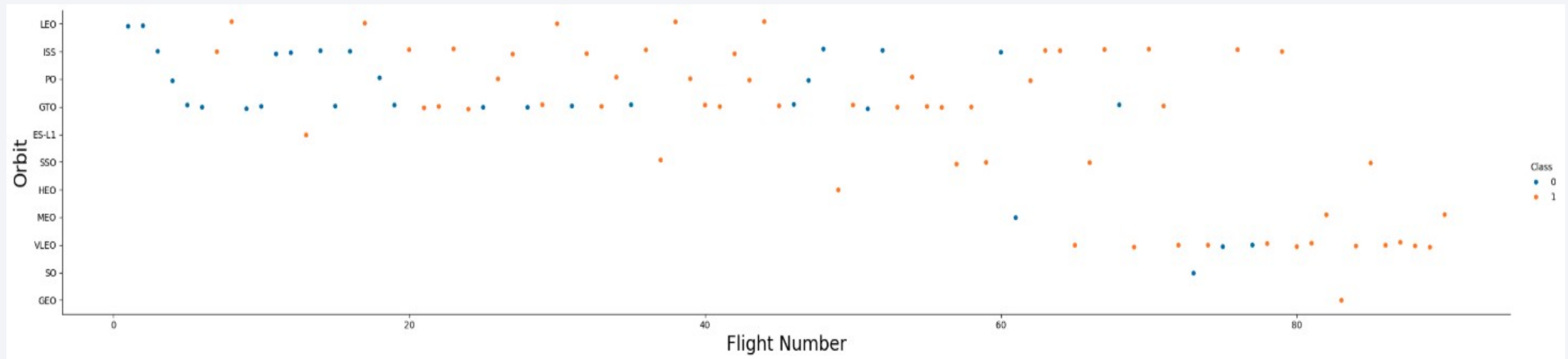
- KSC LC 39A and VAFB SLC 4E launch sites have a high success rate for payloads lower than 6000 Kg.
- CCAFS SLC 40 has a mixed success rate for payloads lower than 6000 Kg and a high success rate for payloads greater than 6000 Kg.

Success Rate vs. Orbit Type



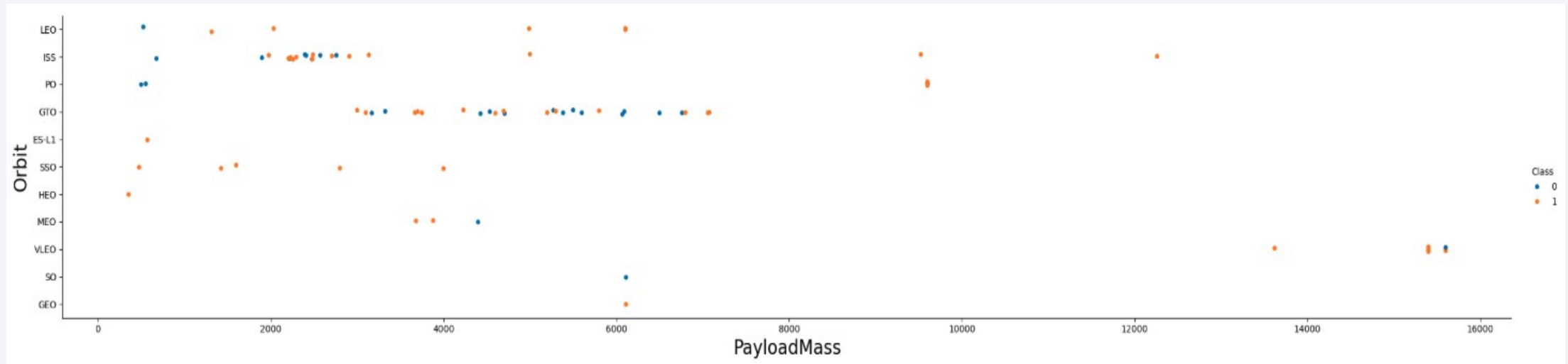
- ES-L1, GEO, HEO and SSO orbits have 100% success rate.
- GTO, ISS, LEO, MEO, PO and VLEO orbits have intermediate success rate (50-85%)
- SO orbit has 0% success rate

Flight Number vs. Orbit Type



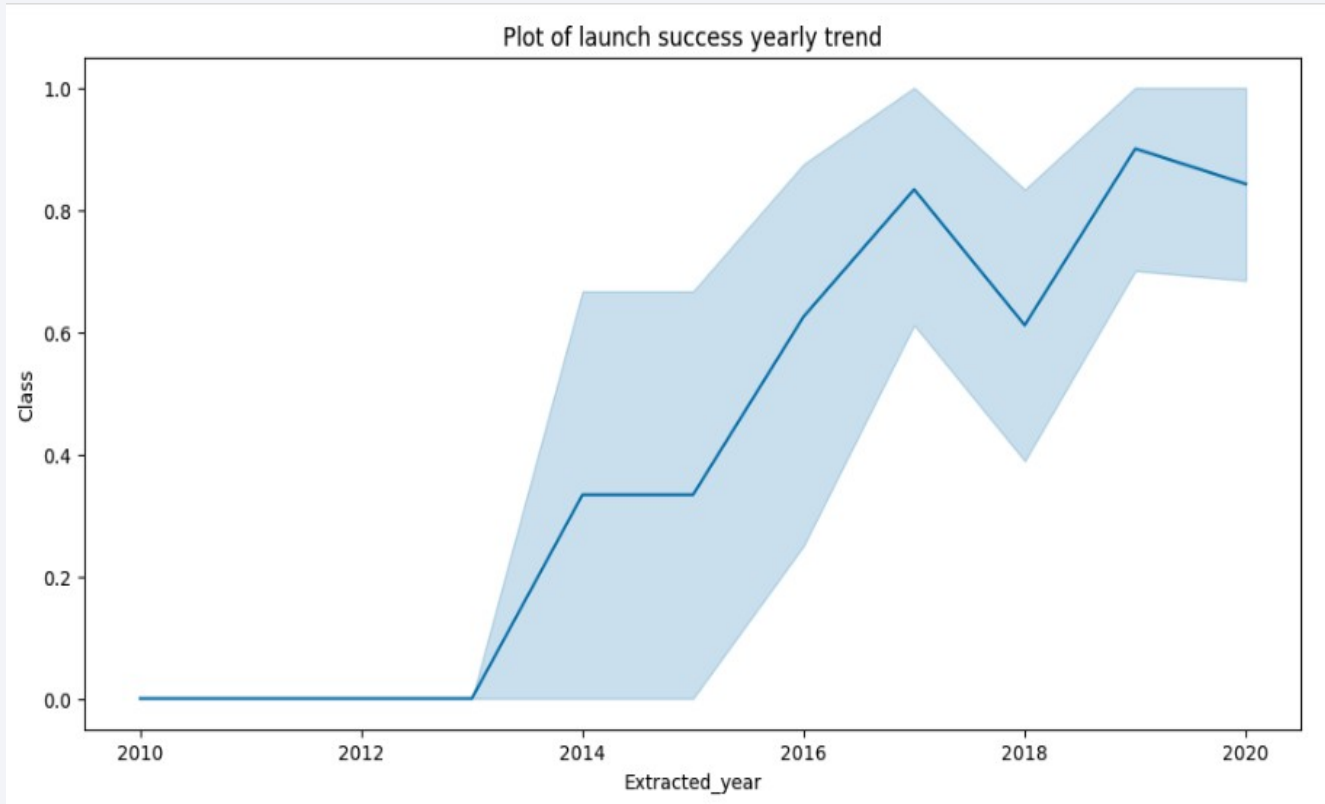
- The 100% success orbits have very few flights (≤ 5)
- In the LEO orbit there seems to be a relationship between success and the number of flight, not so in the other intermediate success-rate orbits.

Payload vs. Orbit Type



- PO and ISS orbits are successful with heavy payloads.

Launch Success Yearly Trend



- The landing success rate has increased from 0% in 2013 up to ~80% by 2020.

All Launch Site Names

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

The keyword DISTINCT is used to retrieve unique values from a specified column or set of columns in a database table.

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2

```
%sql SELECT * FROM SPACEXTABLE  
WHERE Launch_Site LIKE '%CCA%'  
LIMIT 5;
```

This command retrieves the first 5 rows from the table SPACEXTABLE where the Launch_Site column contains the text “CCA” at the beginning of the name.

Total Payload Mass

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer=='NASA (CRS)';  
* sqlite:///my_data1.db  
Done.  


| SUM(PAYLOAD_MASS__KG_) |
|------------------------|
| 45596                  |


```

This command retrieves the sum of the Payload mass carried by boosters launched by NASA (CRS)

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version=='F9 v1.1';
```

```
* sqlite:///my_data1.db  
Done.
```

AVG(PAYLOAD_MASS__KG_)
2928.4

This command retrieves the average payload mass carried by booster version F9 v1.1

First Successful Ground Landing Date

```
%sql SELECT MIN(DATE) FROM SPACEXTABLE WHERE Landing_Outcome LIKE '%SUCCESS%';  
* sqlite:///my_data1.db  
Done.  


| MIN(DATE)  |
|------------|
| 2015-12-22 |


```

This command lists the date when the first successful landing outcome in ground pad was achieved

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT DISTINCT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTABLE  
WHERE Landing_Outcome=='Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000  
AND 6000
```

This command lists the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Booster_Version	PAYLOAD_MASS__KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

Total Number of Successful and Failure Mission Outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) FROM SPACEXTABLE  
GROUP BY Mission_Outcome;
```

This command lists the total number of successful and failure mission outcomes

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

```
%sql SELECT Booster_Version,  
PAYLOAD_MASS__KG_ FROM  
SPACEXTABLE WHERE  
PAYLOAD_MASS__KG_ == (SELECT  
MAX(PAYLOAD_MASS__KG_) FROM  
SPACEXTABLE);
```

This command lists the names of the booster_versions which have carried the maximum payload mass

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

```
%sql SELECT Date,Landing_Outcome,Booster_Version,Launch_Site FROM SPACEXTABLE WHERE  
Landing_Outcome=='Failure (drone ship)' AND Date BETWEEN '2015-01-01' AND '2015-12-31';
```

This command lists the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

Date	Landing_Outcome	Booster_Version	Launch_Site
2015-01-10	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql SELECT Landing_Outcome,  
COUNT(Landing_Outcome) FROM  
SPACEXTABLE WHERE Date BETWEEN  
'2010-06-04' AND '2017-03-20' GROUP BY  
Landing_Outcome ORDER BY  
COUNT(Landing_Outcome) DESC;
```

This command ranks the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Landing_Outcome	COUNT(Landing_Outcome)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left portion shows a clear blue sky.

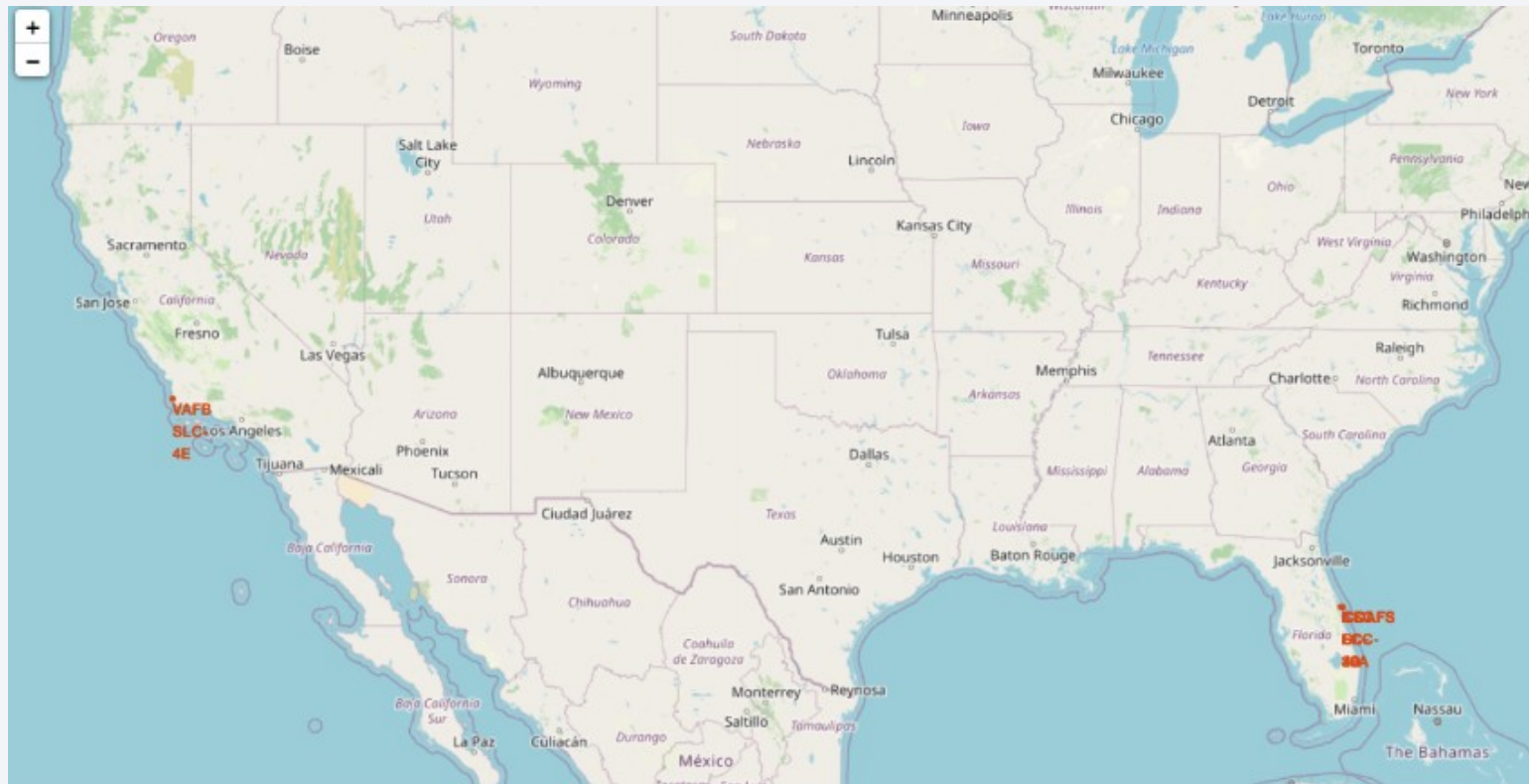
Section 3

Launch Sites Proximities Analysis

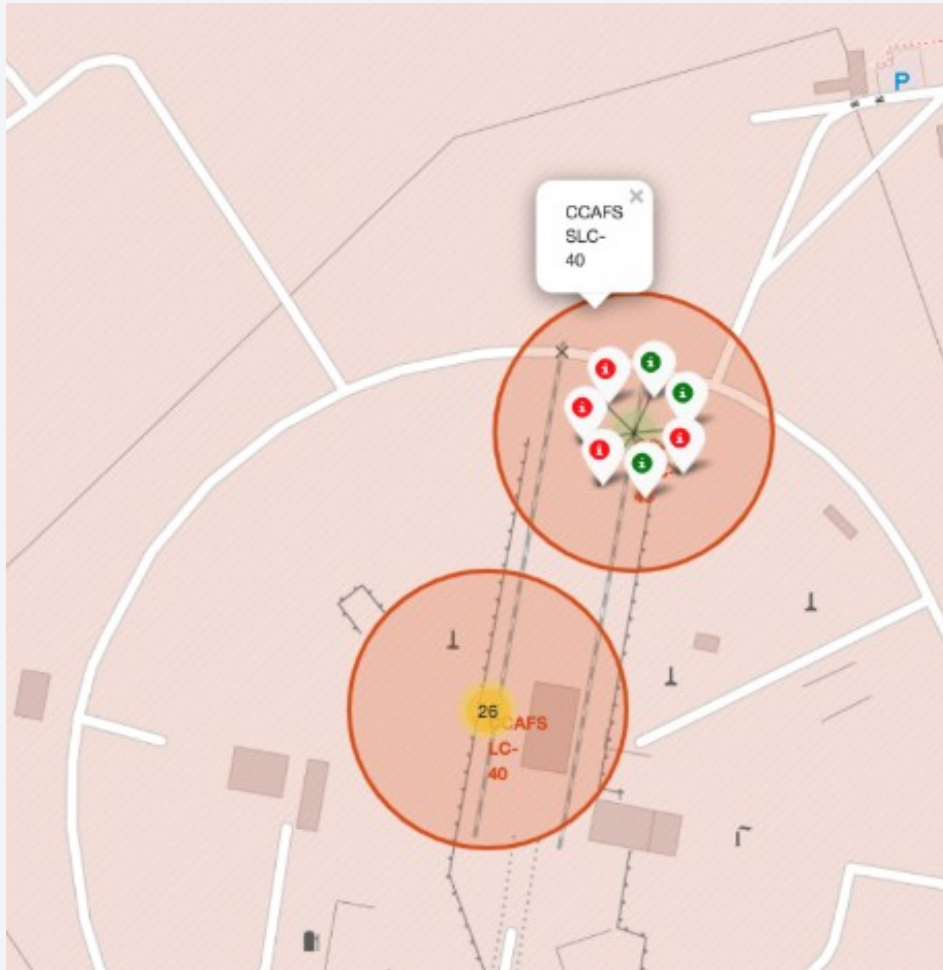
All launch sites' location markers

SpaceX launch sites are in the United States of America.

They are located near the coasts to launch rockets towards the ocean and minimize debris falling on populated areas.



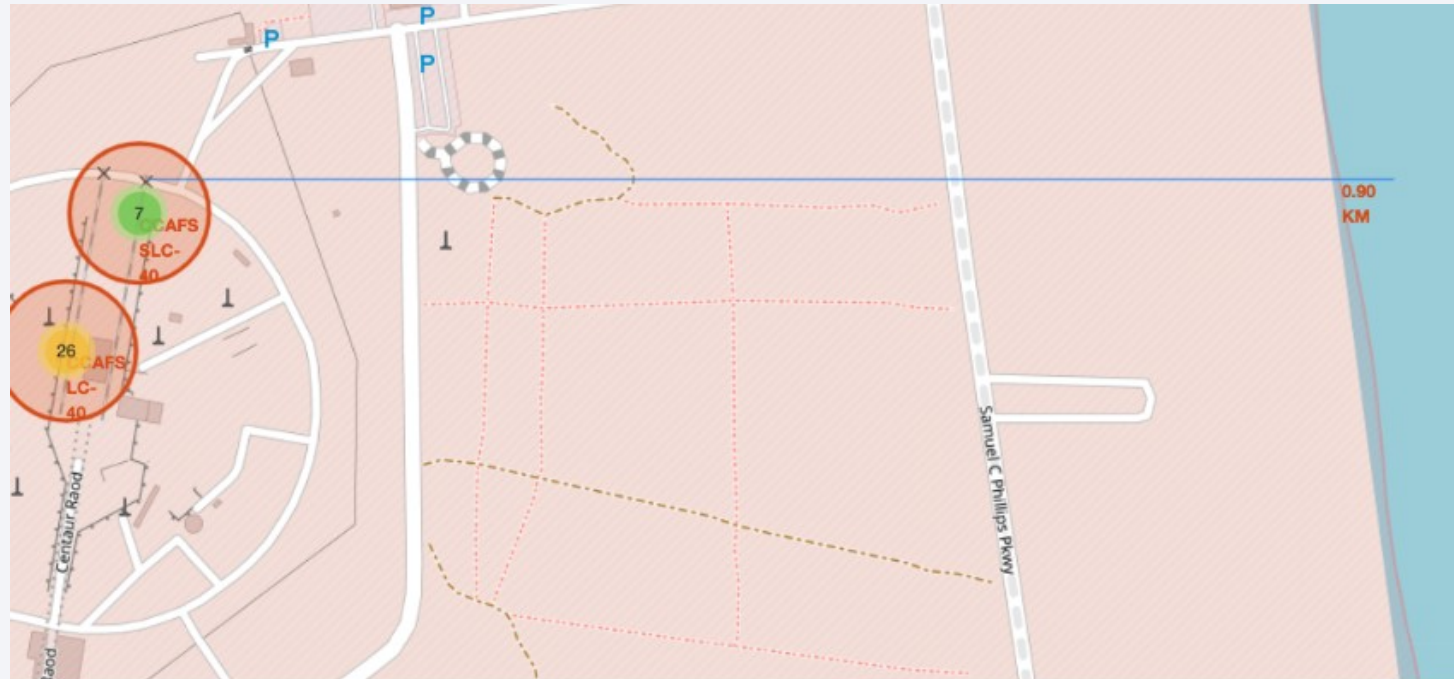
Color-labeled launch records on the map



Colour labeled markers help identify launch sites with high success rate:

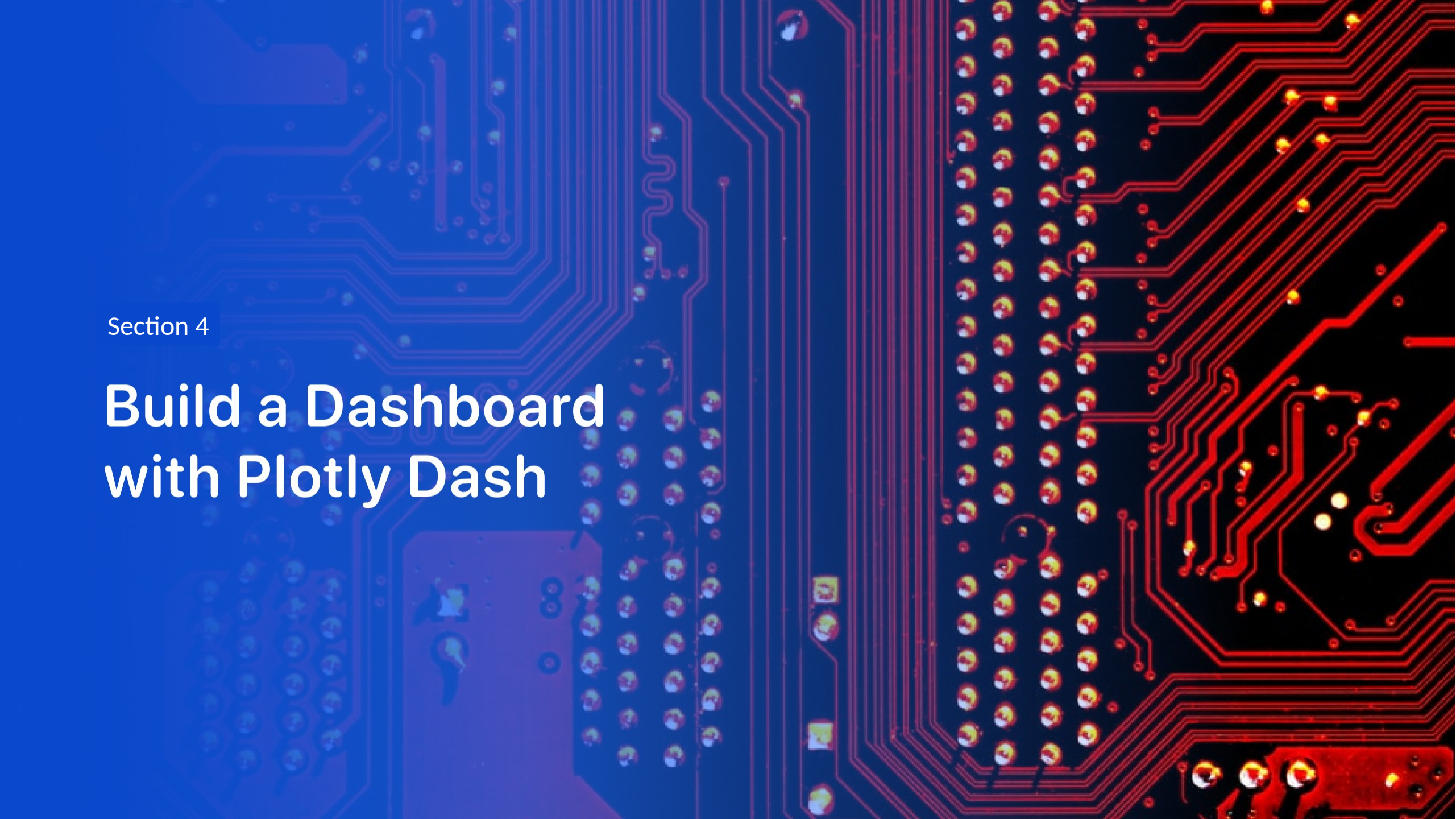
- Green: Successful launch.
- Red: Failed launch.

Distance from the launch site CCAF SLC-40 to the coast line



With the launch sites displayed on the map it was possible to observe that launch sites

- Are in close proximity to railways, highways and coastlines.
- Are not in close proximity to cities to reduce the risk to the population.

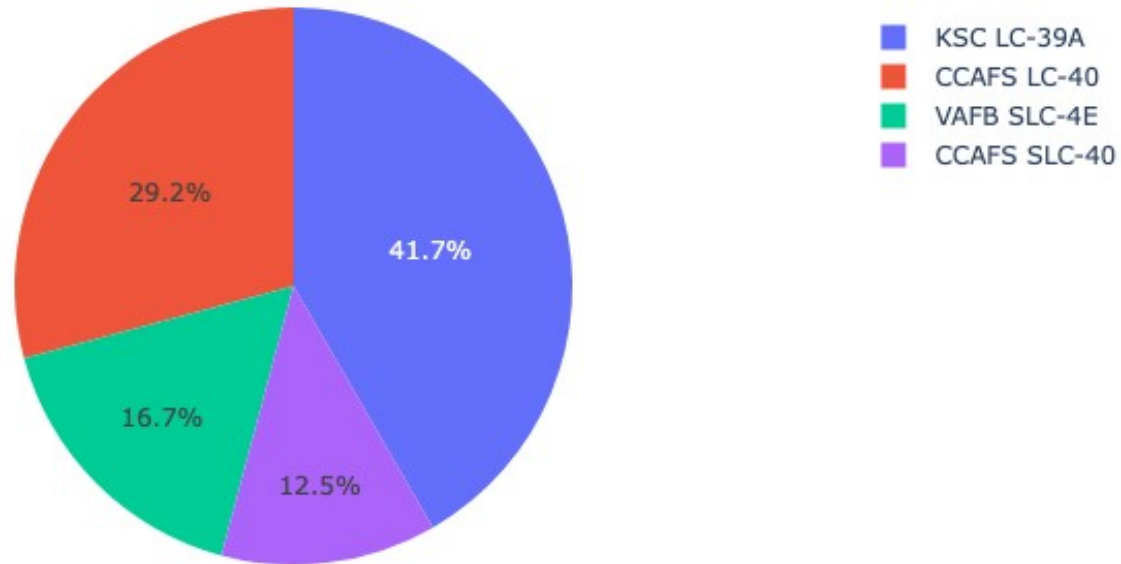


Section 4

Build a Dashboard with Plotly Dash

Contribution to **successful** launches per launch site

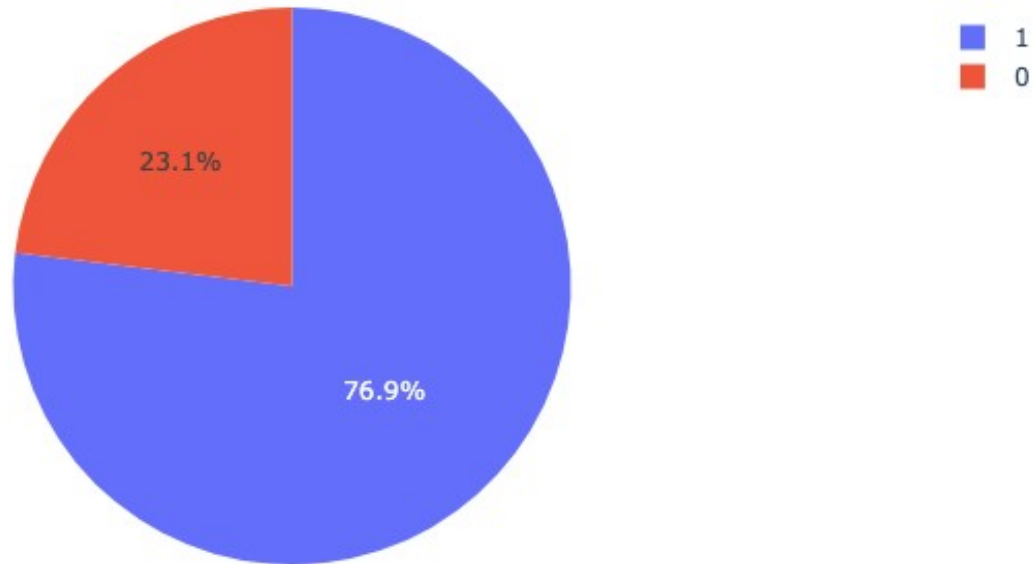
Success Count for all launch sites



KSC LC-39A has the highest number of successful launches

Percentage of success at KSC LC-39A

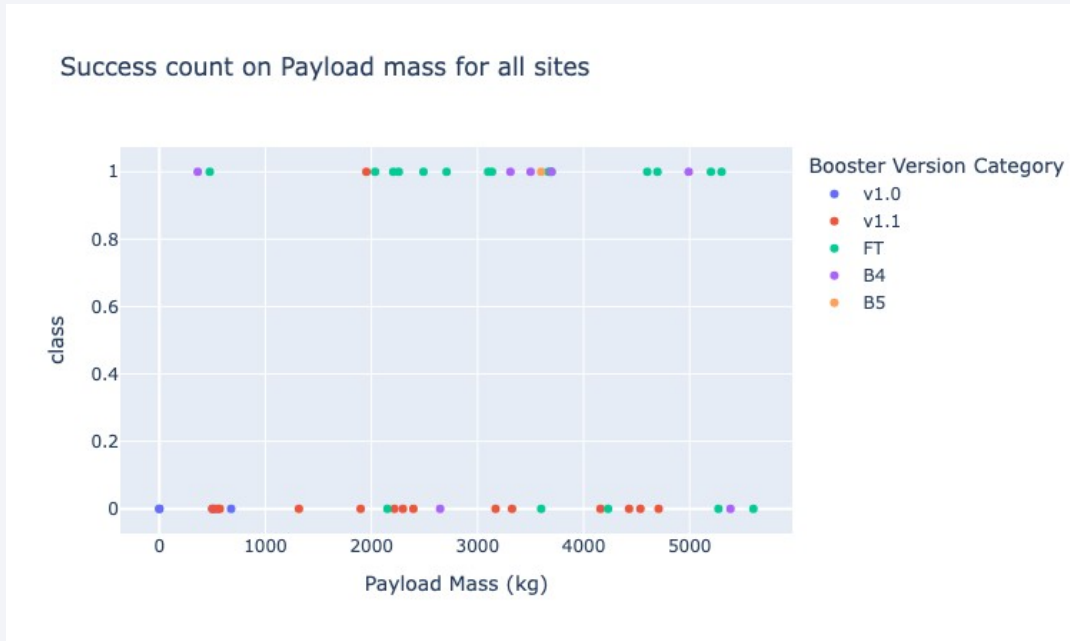
Total Success Launches for site KSC LC-39A



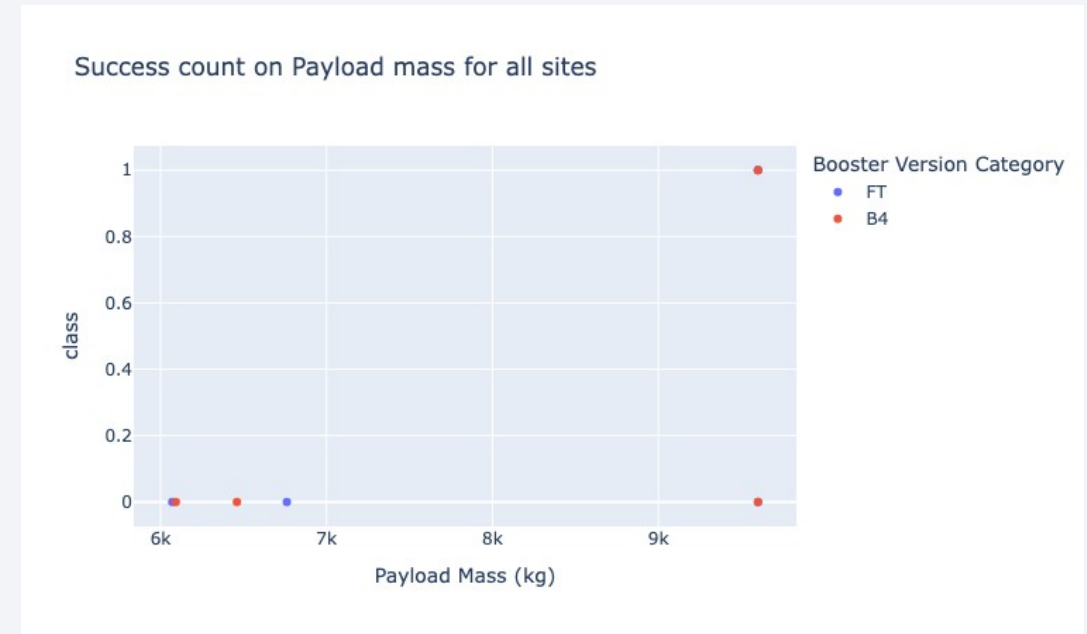
KSC LC-39A has 76.9% of success rate

Outcome vs different ranges of Payload

Payload 0 – 6000 Kg



Payload 6000 - 10000 Kg



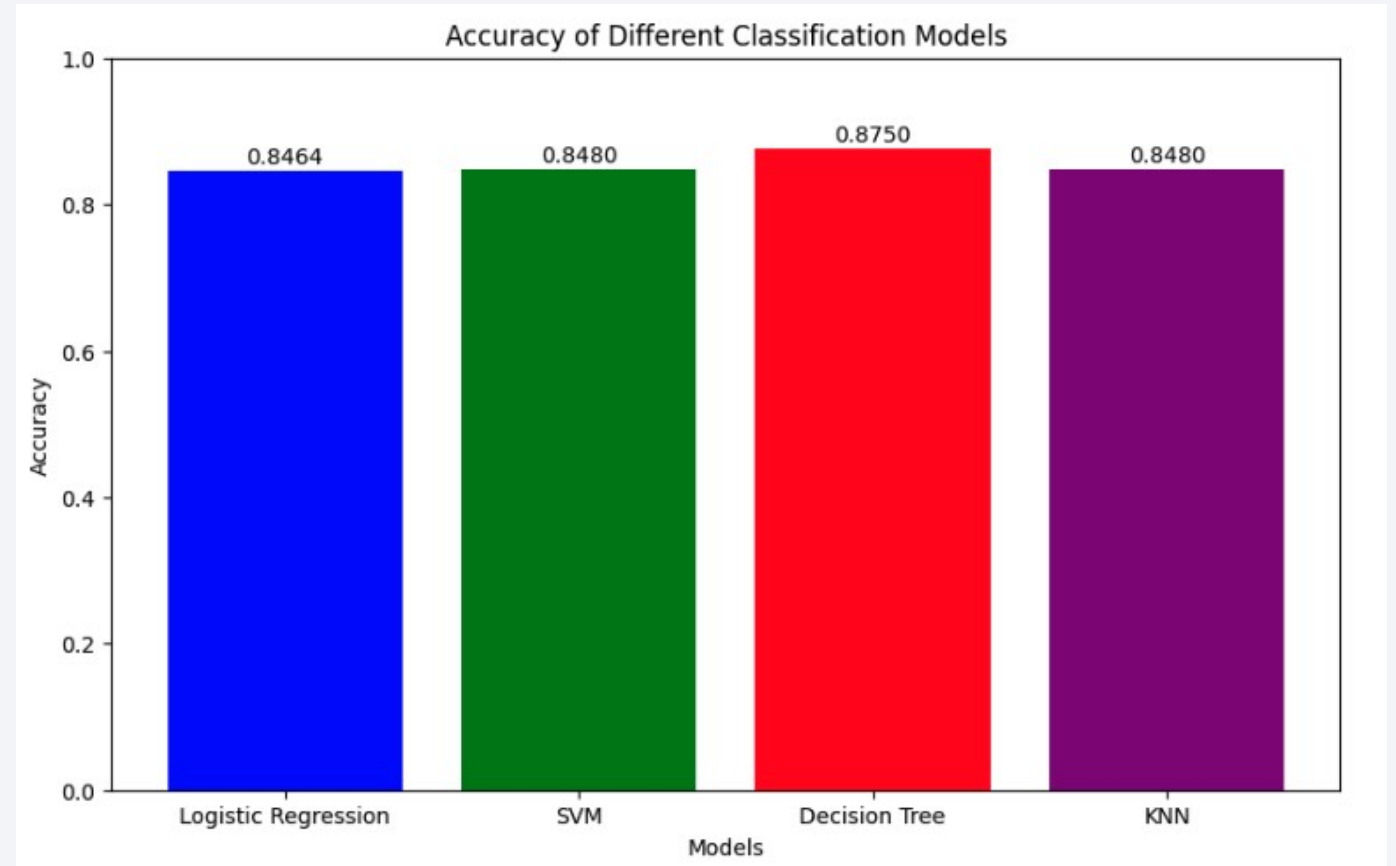
The success rate is lower for heavy payloads (>6000 Kg)

Section 5

Predictive Analysis (Classification)

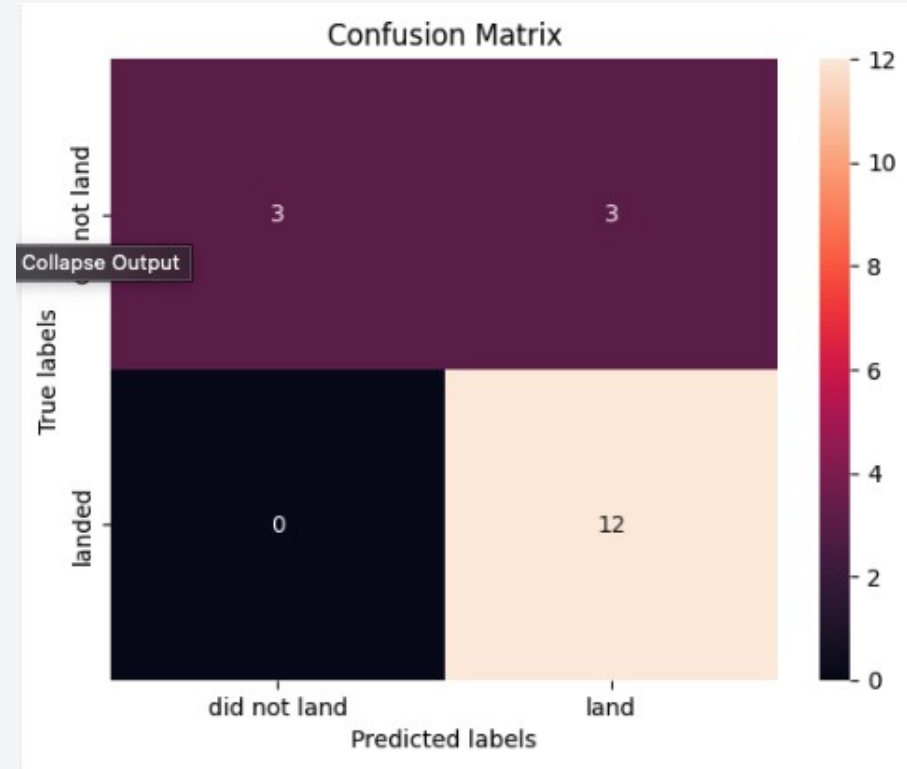
Classification Accuracy

- The Decision Tree model seems to perform better on the validation data set based on the evaluation of the accuracy metric.



Confusion Matrix

- All the models have virtually the same confusion matrix on the test dataset. Their main problem is the false positives.



Whole data set metric

- In this scenario, Precision for the 'successful landing' is the most important evaluation metric because false positives (predicting a success when the landing actually fails) are the most costly errors.
- Metrics for the whole data set were requested to choose the best method.
- According to the metrics of the whole dataset, the best performing model is the SVM.

	LogReg	SVM	Tree	KNN
Precision_Score	0.833333	0.845070	0.830986	0.830986
F1_Score	0.909091	0.916031	0.900763	0.900763
Accuracy	0.866667	0.877778	0.855556	0.855556

Conclusions

- Low to moderate Payloads (>0 – 5000 Kg) have better landing success rate than heavy (>6000 Kg) Payloads.
- The Support Vector Machine model built in this project can be used to predict with high Precision (0.845) the successful landing of future launches.
- As more data becomes available it could be included in the dataset to improve the metrics of the models.

Appendix

GitHub Repository URL:

https://github.com/morca76/DS_CapstoneProject

Thank you!

